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Information Engineering

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IDEAL-RAG: 用於檢索增強生成的指令驅動雙觀點萃取與對齊連結方法

**IDEAL-RAG: Instruction-driven Dual-standpoint Elicitation
and Alignment Linking for Retrieval Augmented Generation**

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摘要

檢索增強生成 (RAG) 讓大型語言模型 (LLM) 能即時引用外部證據，但只要檢索結果摻雜雜訊或對抗片段時，主流方法往往拋棄自身參數知識，直接複述錯誤內容並引發幻覺。既有工作雖提出各自策略解決雜訊問題，整體卻仍缺乏對如何系統性調度與對齊內外知識的深入探討；近期機制層研究更發現，面對強污染檢索，LLM 依賴內部記憶的程度急劇下降。

為此，我們提出 **IDEAL-RAG**，一個三階段指令驅動式框架，讓模型先顯式喚醒自身記憶，再讓內、外兩源各自生成自洽論點，最後在鏈結模組對證據交叉驗證並產生可追蹤推理，全程無需改動檢索器或額外人工標註。此外，我們提供一套機制層面診斷方法：以自建的對抗敏感度指標 (CSS) 搭配現有的分層參數知識分數 (PKS)，觀察 RAG 在強污染環境下「知識-FFN」通道的動態，並驗證「預先提取並平衡殘差」的步驟確實能抑制內知失衡、降低幻覺風險。

在實驗中表明，IDEAL-RAG 在乾淨檢索下，與現有的強勁基準方法 InstructRAG 相當；於對抗式強污染測試集中，則準確率最高提升 +22.8%，準確率衰減率減半。透過機制分析進一步證實，本框架可穩定調用內知並維持答案置信度。結果證實：在 LLM「已知」與「新讀」之間先立論點、再理性對照，是構築更可靠 RAG 系統的有效途徑。

關鍵字：大型語言模型, 檢索增強生成, 上下文學習, 問答系統, 穩健自然語言處理, 幻覺降低, 提示工程

Abstract

Retrieval-augmented generation (RAG) lets large language models (LLMs) cite fresh external evidence on demand. Yet when the retrieved texts contain noise or adversarial edits, most RAG systems ignore their parametric knowledge and simply echo the wrong passages, producing hallucinations. Existing studies propose various denoising strategies, but few explore how to systematically balance and align internal and external knowledge. Recent probing work even shows that, under heavy corruption, an LLM’s reliance on its stored memory drops sharply.

We introduce **IDEAL-RAG**, a three-stage, instruction-driven framework. The model (i) explicitly recalls its latent knowledge, (ii) lets the internal and retrieved sources each craft a self-contained standpoint, and (iii) cross-checks these standpoints in a linking module to yield a traceable rationale—all without changing the retriever or adding manual labels. To inspect what happens inside the network, we pair a new Counterfactual Sensitivity Score (CSS) with the existing layer-wise Parametric Knowledge Score (PKS), revealing how the “knowledge-FFN” pathway behaves under strong noise and showing that our “extract-then-balance” step reduces hallucination risk.

Experiments demonstrate that IDEAL-RAG matches the strong InstructRAG baseline on clean retrieval, while under adversarially corrupted contexts, it boosts exact-match accuracy by up to +22.8 % and cuts accuracy loss by half. CSS and PKS analyses confirm that the framework keeps answer confidence steady by drawing on internal facts when external evidence is unreliable. These findings indicate that first establishing separate standpoints and then reasoning over their agreement is an effective route to more dependable RAG systems.

Keyword: Large Language Models, Retrieval-Augmented Generation, In-Context Learning, Question Answering, Robust NLP Systems, Hallucination Reduction, Prompt Engineering

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Chapter 1

Introduction

1.1 Background

Large language models (LLMs) have demonstrated remarkable capabilities in natural language generation[3, 39, 41, 4]. However, when faced with questions about recent events, rare entities, or domain-specific facts that were not encountered during pre-training, they often fail to respond accurately, leading to factual hallucinations or outright refusals to answer[34, 8, 20, 55, 57, 47]. Continuously updating model parameters or training LLMs to cover all knowledge domains is impractical and resource-intensive.

In open-domain question answering (ODQA)[5], *retrieval-augmented generation* (RAG, illustrated in Figure 1.1) has emerged as a mainstream solution[10, 11, 18, 23, 26]. By prepending retrieved passages from a large corpus to the input query, a language model is expected to ground its answer in verifiable, up-to-date evidence, while still leveraging its vast **parametric knowledge**.

This idealized setup rests on two key assumptions:

1. the retrieved documents are accurate, relevant, and self-consistent; and
2. the model can rationally balance external information against its internal knowledge.

Recent studies have shown that these assumptions often break down in real-world scenarios.

1.2 Motivation

Retrieval pipelines frequently return *noisy* content, including irrelevant snippets, partial matches, and even adversarially inserted misinformation, collectively referred to as **RAG noise**[9, 52, 54, 27, 7, 46]. In such cases, state-of-the-art RAG systems exhibit a concerning failure

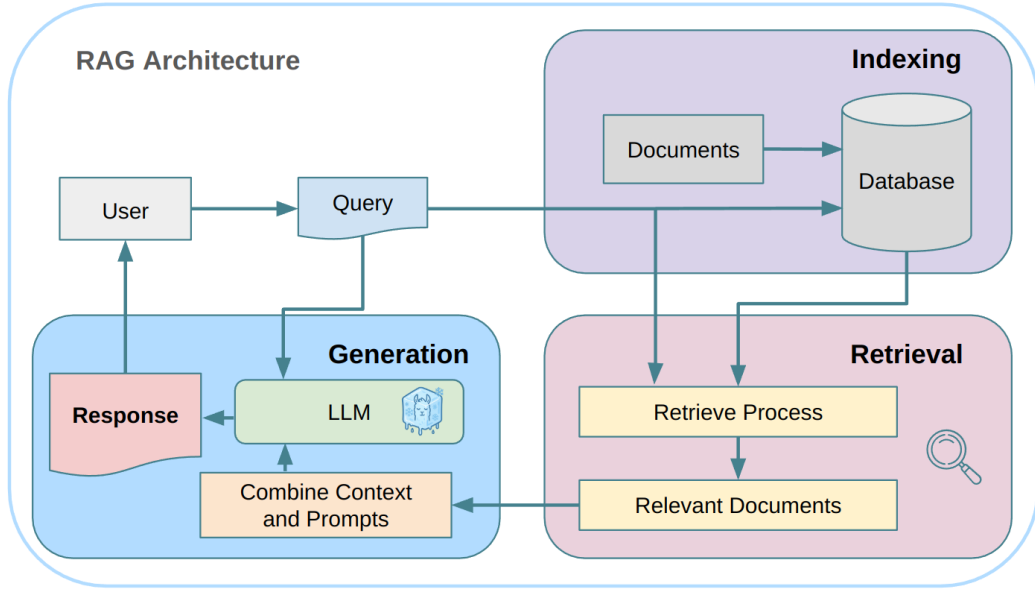


Figure 1.1: A representative instance of the RAG process for question answering. Reproduced from [10].

mode: **LLMs tend to over-rely on the retrieved context while neglecting their parametric knowledge**[42, 38]. A single manipulated sentence can lead the model to produce incorrect answers confidently.

To address this issue, recent work has proposed strategies for enhancing the robustness of RAG under noisy conditions, aiming to help models resist being misled[9, 56]. However, these methods often lack transparency or interpretability, leaving the model’s decision-making process hard to trace.

Meanwhile, a complementary line of research explores when and if retrieval should be used at all, suggesting that models can enhance noise robustness by falling back on parametric memory when external evidence appears unreliable. Approaches include trigger-based or confidence-aware controllers that disable retrieval in low-trust settings [52, 45], and hybrid schemes that dynamically down-weight or reject passages conflicting with high-confidence internal knowledge [54, 48, 1].

Parallel research has begun to explore the conflict between external and internal knowledge within RAG workflows[50, 43]. Studies have observed that current models disproportionately prioritize retrieved content, even when it conflicts with memorized facts. Although some approaches attempt to increase the influence of parametric knowledge, they typically

rely on shallow prompting strategies such as: *”If the documents are insufficient, answer from your own knowledge.”*

Yet, such prompts offer no concrete mechanism to detect insufficiency, resolve conflicting signals, or motivate trust in internal memory. Empirical findings consistently reveal that LLMs continue to echo superficially plausible but incorrect evidence naively.

1.3 Our Work

In short, the field still lacks an answer to a core question:

How can we leverage what the model already “knows” to improve robustness when retrieval is incomplete or misleading?

To fill this gap, we propose **IDEAL-RAG (Instruction-driven Dual-standpoint Elicitation and Alignment Linking)**—a structured framework that systematically addresses three core components:

1. **explicit elicitation of parametric knowledge;**
2. **dual-source reasoning via internal and external standpoints;** and
3. **stable integration into a final unified rationale.**

Our framework first instructs the model to generate its own internal knowledge[43, 23], then separately derives internal and external *standpoints* before performing a deliberate integration step. This pipeline encourages a deliberative process that respects both retrieved evidence and intrinsic memory, as shown in Figure 1.2.

Empirical results across multiple datasets demonstrate that IDEAL-RAG is more robust under counterfactual or partially corrupted retrieval settings. Moreover, in cases where internal and external knowledge are complementary, our approach yields greater stability and maintains strong performance even when the retrieval is clean.

In summary, the results and contribution of our work are as follows:

1. We introduce **IDEAL-RAG**, an instruction-guided framework that explicitly separates and reconciles parametric and retrieved evidence—effective in both training-free and

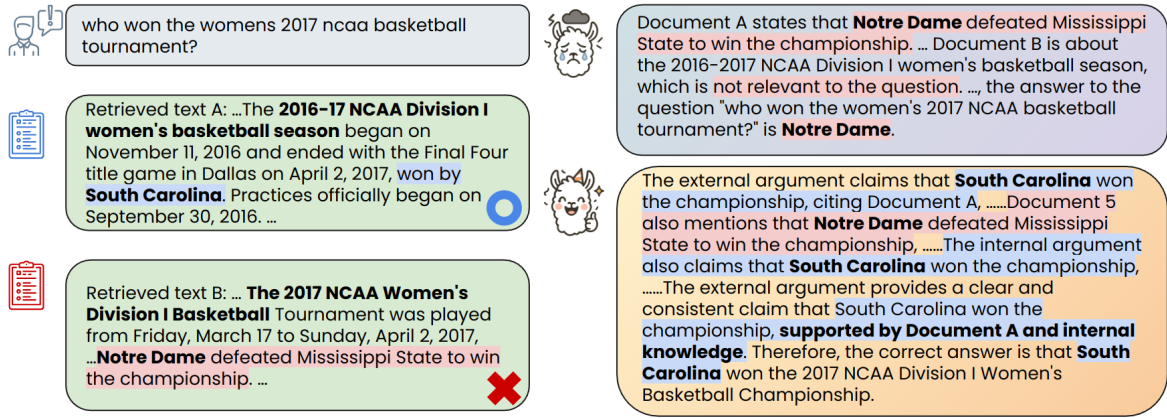


Figure 1.2: The figure contrasts a standard retrieval-augmented baseline with IDEAL-RAG under a noisy-retrieval scenario. Whereas the baseline model (upper-right) anchors on misleading context and produces an incorrect rationale, IDEAL-RAG (lower-right) draws on its internal knowledge, balances conflicting sources, and delivers a stable, correct answer, demonstrating its robustness to retrieval noise.

fine-tuned settings.

2. Across four open-domain QA benchmarks and two counterfactual noise regimes, IDEAL-RAG sacrifices a small amount of clean-data accuracy compared to standard RAG, while **demonstrating much stronger robustness**.
3. Our results suggest that **reasoned negotiation between what an LLM already “knows” and what it newly “reads” is a principled path toward dependable retrieval-augmented generation**.

Chapter 2

Related Work

2.1 Noise Sensitivity in Retrieval-Augmented Generation

Early RAG systems implicitly assume that retrieved passages are **accurate, mutually consistent and fully supportive** of the target answer. Surveys on retrieval-augmented generation and on LLMs for information retrieval [10, 58] quickly overturned this assumption, showing that even minimal noise or adversarial edits can flip a correct answer into a confident hallucination. This observation has spawned a distinct research line on *noise-robust RAG*.

Several studies investigate how different types of noise degrade an LLM’s behaviour [9, 56, 6, 54, 7]. To mitigate such degradation, recent work explores diverse defences.

- Yu, Wenhao et al. [54] forced the model to justify each passage before answering.
- Xiang, Chong et al.[48] aggregated answers from multiple document subsets to dilute spurious evidence.
- Asai, Akari et al. [1] coupled reflection with RLHF scoring [36, 31], generating multiple reasoning chains and using self-consistency voting to drop contradictory paths; its gains are bounded by chain quality and voting thresholds.
- Further studies on model knowledge and the distractibility of LLMs [40, 37] underscored the severity of this issue.
- Wei, Zhepei et al. [45] followed a streamlined two-phase procedure, called InstructRAG. First, in a small answer-seen stage, a handful of handcrafted exemplars teach a frozen LLM how to transform a set of retrieved passages into an explicit, step-by-step rationale that reaches the gold answer. The same prompt template is then applied in the answer-unseen test phase.

Because it requires no additional supervision signals, re-rankers, or auxiliary models, the entire pipeline fits inside a single prompt and can be run on modest hardware—hence its reputation for being lightweight. Despite this simplicity, InstructRAG attains near-state-of-the-art exact-match scores on several QA benchmarks.

Subsequent analyses, however, reveal a vulnerability: if even one retrieved sentence is adversarially edited, the model often reproduces the corruption verbatim, showing that instruction alone cannot guarantee a balanced use of retrieved evidence versus parametric memory [38].

Given its strong baseline accuracy, minimal engineering overhead, and well-documented susceptibility to noisy retrieval, **InstructRAG serves as the primary comparison point in all of our experiments.**

2.2 Adversarial and Counterfactual Benchmarks

To quantify RAG brittleness, a series of controlled benchmarks inject artificial noise into the retrieval pipeline.

- Niu, Cheng et al. [47] compiled a corpus for token-level hallucination analysis across domains. Yoran, Ori et al. [52] randomly inserted irrelevant or misleading sentences into HotpotQA[51] and NQ[24], and compare NLI-filtered defences [14, 2].
- Zhang, Tianjun et al. [56] offered paragraph-level replacement, deletion, and insertion attacks, and even forced the model to answer when the retrieved set contains no gold evidence.
- Fang, Feiteng et al. [9] defined three noise classes—(i) *relevant retrieval noise* (retrieved but answer-free), (ii) *irrelevant retrieval noise* (random passages) and (iii) *counterfactual noise* (factually incorrect content). It then applied adaptive adversarial training with multiple loss terms.

Building on this line of work, we create a counterfactual test set to measure a model’s reliance on external evidence and its resistance to polluted contexts. Our experiments show that, **after integrating internal and external knowledge, the model remains stable even under more complex retrieval scenarios.**

2.3 Parametric Knowledge in RAG

- Relatively few studies have examined how an LLM’s **parametric memory** contributes to RAG. Work on *conditional RAG* [49, 30, 19, 1] asked when retrieval should be invoked to avoid distracting the model when its own memory suffices, but offered no mechanism for mutual verification.
- Wang, Fei et al. [43] first prompted the model to generate its own memory about the question, then clustered the retrieved evidence with the extracted knowledge for post-hoc fusion. However, their method remains a prompt-level heuristic without a learnable mechanism for resolving conflicts.
- Sun, Zhongxiang et al. [38] probed the internal mechanism by proposing Parametric Knowledge Score (PKS) and External Context Score (ECS), and found that mainstream RAG architectures rely almost exclusively on retrieval.

We extend analysis from [38]: besides **PKS** we design a **Counterfactual Sensitivity Score** (CSS) to quantify how internal knowledge buffers against noisy evidence. Our results confirm that **explicitly extracting parametric knowledge** and **integrating it through dual-source deliberation** significantly improves robustness.

Chapter 3

Methodology

Large language models (LLMs) have shown strong ability to follow complex instructions, preserve style, and produce multi-step explanations with minimal supervision. Recent work [3, 1, 45] indicates that **with carefully curated exemplars, existing models can acquire sophisticated behaviors without heavy annotation or custom rewards**. Building on these insights, we introduce IDEAL-RAG (Instruction-Driven Evidence Alignment and Linking), a three-stage, instruction-centric pipeline that first exposes and contrasts the LLM’s parametric memory with potentially noisy retrieved passages, then reconciles them.

Why three stages? Prior RAG methods often treat internal knowledge as a mere fallback. In practice, deciding *when* to trust memory versus retrieved text is non-trivial, and modern LLMs may overweight superficially plausible passages—especially under partial or adversarial retrieval. While some approaches suppress parametric knowledge to rely almost entirely on external evidence, real deployments cannot assume perfect retrieval. We take the opposite view: *proactively* surface what the model already “knows”, require two independent standpoints—one from parametric memory and one from retrieved context—and then introduce a *linking* step that explicitly reconciles them. This separation forces the model to reason over both sources rather than latching onto the most salient passage, yielding a more transparent and robust route to the final answer.

The pipeline operates in two mirrored regimes (Figure 3.1). On a small answer-seen split, we produce gold-guided standpoints and a linked rationale; these outputs form a compact demo bank capturing ideal reasoning trajectories. For answer-unseen questions, the model receives the query, retrieved passages, and extracted internal knowledge and—conditioned only on the demos—generates standpoints and the final linked rationale. The linking step compels an explicit comparison and reconciliation of the standpoints.

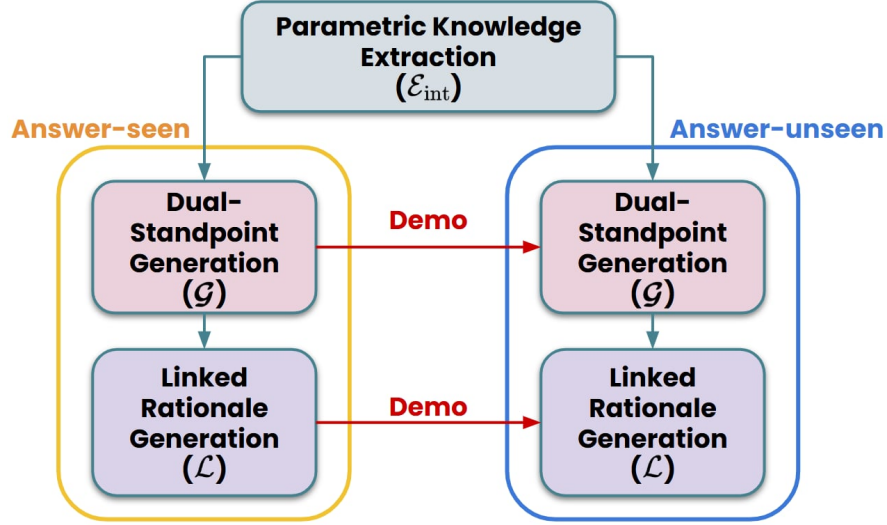


Figure 3.1: High-level view of IDEAL-RAG. The pipeline first elicits parametric knowledge ($\mathcal{E}_{\text{int}}$), then runs two stages in mirrored regimes. On a small answer-seen split, it generates internal/external standpoints ($\mathcal{G}$) and a linked rationale ($\mathcal{L}$) to build a demo bank. On answer-unseen questions, the same two stages are executed conditioned on these demos, yielding the final linked rationale and answer. This layout makes internal and retrieved evidence explicit and comparable before integration.

With this overview in place, we now detail the three modules:

1. **Parametric Knowledge Extraction ($\mathcal{E}_{\text{int}}$)**
2. **Dual-Source Standpoint Generation ($\mathcal{G}$)**
3. **Linked Rationale Generation ($\mathcal{L}$)**

All three stages **operate solely on the frozen backbone** Θ_0 , requiring neither teacher models nor external verifiers. Ground-truth answers are revealed only when constructing the exemplar bank $\mathcal{B}$, allowing the model to **observe ideal reasoning trajectories** before engaging with unseen questions via few-shot (ICL) or lightweight instruction tuning (SFT). An overview and detailed procedure are provided in Algorithm 1 and Figure 3.2.

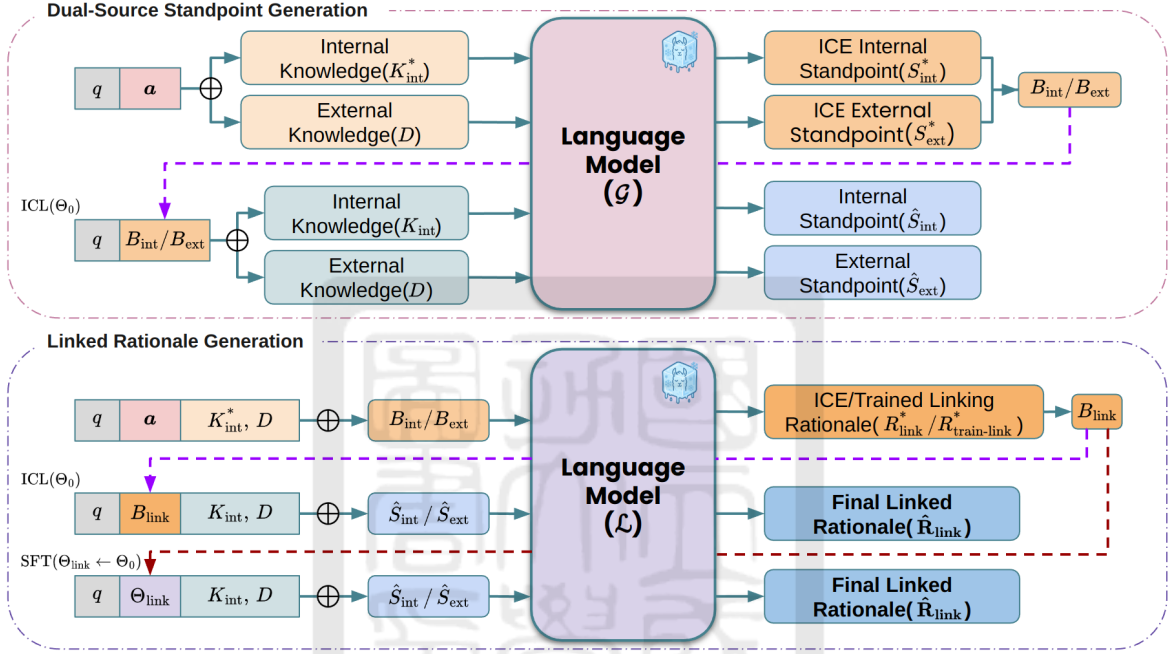


Figure 3.2: Overview of IDEAL-RAG. The system first constructs two standpoints: one derived from the model’s internal memory, and the other from the retrieved passages. These standpoints serve as an interface that bridges parametric and non-parametric knowledge. The model then performs deliberative reasoning over the two to generate a unified rationale and final answer. This reasoning is implemented either via in-context prompting or through a lightweight fine-tuned prediction head. The full process comprises two stages: one where the answer is revealed (answer-seen) and one where it remains hidden (answer-unseen).

Algorithm 1 IDEAL-RAG

Require: QA corpus $\mathcal{C} = \{(q_i, a_i, \mathcal{D}_i)\}$ (including train and test); frozen backbone Θ_0

Ensure: Fusion weights Θ_{fuse} ; at test time $\hat{\mathcal{R}}_{\text{fuse}}$

Standpoint Generation ($k \ll |\mathcal{C}|$):

- 1: **for all** $(q, a, \mathcal{D}) \in \mathcal{C}_{\text{seed}}$ **do**
- 2: $K_{\text{int}}^* \leftarrow \mathcal{E}_{\text{int}}(q; \Theta_0)$ ▷ Extracted Parametric Knowledge
- 3: $\mathcal{S}_{\text{int}}^* \leftarrow \mathcal{G}(q, a, K_{\text{int}}^*; \Theta_0)$ ▷ answer-seen
- 4: $\mathcal{S}_{\text{ext}}^* \leftarrow \mathcal{G}(q, a, \mathcal{D}; \Theta_0)$
- 5: add $\mathcal{S}_{\text{int}}^*$ to internal exemplar bank $\mathcal{B}_{\text{int}}$
- 6: add $\mathcal{S}_{\text{ext}}^*$ to external exemplar bank $\mathcal{B}_{\text{ext}}$
- 7: **for all** $(q, a, \mathcal{D}) \in \mathcal{C} \setminus \mathcal{C}_{\text{seed}}$ **do** ▷ Standpoint Generator (both test and train)
- 8: $K_{\text{int}} \leftarrow \mathcal{E}_{\text{int}}(q; \Theta_0)$ ▷ answer-unseen
- 9: $\hat{\mathcal{S}}_{\text{int}} \leftarrow \mathcal{G}(q, K_{\text{int}}; \Theta_0, \text{ICE} = \mathcal{B}_{\text{int}})$
- 10: $\hat{\mathcal{S}}_{\text{ext}} \leftarrow \mathcal{G}(q, \mathcal{D}; \Theta_0, \text{ICE} = \mathcal{B}_{\text{ext}})$

Linked Rationale Generation:

- 1: **for all** $(q, a, \mathcal{D}) \in \mathcal{C}_{\text{seed}}$ **do**
 - 2: $\mathcal{R}_{\text{link}}^* \leftarrow \mathcal{L}(q, a, \mathcal{D}, K_{\text{int}}^*, \mathcal{S}_{\text{int}}^*, \mathcal{S}_{\text{ext}}^*; \Theta_0)$ ▷ answer-seen
 - 3: add $\mathcal{R}_{\text{link}}^*$ to integrated exemplar bank $\mathcal{B}_{\text{link}}$
 - 4: **for all** $(q, a, \mathcal{D}) \in \mathcal{C} \setminus \mathcal{C}_{\text{seed}}$ **do** ▷ Linking-based Integrator
 - 5: **if** mode==In-Context Learning **then**
 - 6: $\hat{\mathcal{R}}_{\text{link}} \leftarrow \mathcal{L}((q, \mathcal{D}, K_{\text{int}}, \hat{\mathcal{S}}_{\text{int}}, \hat{\mathcal{S}}_{\text{ext}}) \in \mathcal{C}_{\text{test}}; \Theta_0, \text{ICE} = \mathcal{B}_{\text{link}})$ ▷ answer-unseen
 - 7: **else if** mode==Fine-Tuning **then**
 - 8: $\mathcal{R}_{\text{train-link}}^* \leftarrow \mathcal{L}((q, a, \mathcal{D}, K_{\text{int}}, \hat{\mathcal{S}}_{\text{int}}, \hat{\mathcal{S}}_{\text{ext}}) \in \mathcal{C}_{\text{train}}; \Theta_0)$ ▷ answer-seen
 - 9: $\Theta_{\text{link}} \leftarrow \text{Update}(\mathcal{R}_{\text{link-train}}^* | ((q, \mathcal{D}, \hat{\mathcal{S}}_{\text{int}}, \hat{\mathcal{S}}_{\text{ext}}) \in \mathcal{C}_{\text{train}}; \Theta_0))$
 - 10: $\hat{\mathcal{R}}_{\text{link}} \leftarrow \mathcal{L}((q, \mathcal{D}, K_{\text{int}}, \hat{\mathcal{S}}_{\text{int}}, \hat{\mathcal{S}}_{\text{ext}}) \in \mathcal{C}_{\text{test}}; \Theta_{\text{link}})$ ▷ answer-unseen
 - 11: **return** $\hat{\mathcal{R}}_{\text{link}}$
-

3.1 Problem Formulation

Consider an open-domain corpus

$$\mathcal{C} = \{(q_i, a_i, \mathcal{D}_i)\}_{i=1}^N, \quad (3.1)$$

where q_i is a natural-language question, a_i its gold answer, and $\mathcal{D}_i = \mathcal{R}(q_i) \subset \mathcal{T}$ a set of passages retrieved from a large text collection $\mathcal{T}$ by an immutable retriever $\mathcal{R}$. In a *textbook* RAG system, one concatenates $[q_i; \mathcal{D}_i]$ and feeds the result into an LLM, hoping the model will ground its answer in $\mathcal{D}_i$ while drawing on any complementary parametric knowledge.

We deliberately freeze the retriever to isolate generation-side robustness. The objective is to maximise standard exact-match accuracy

$$\text{Acc} = \frac{1}{|\mathcal{C}_{\text{test}}|} \sum_{(q,a) \in \mathcal{C}_{\text{test}}} \mathbf{1}[a \subseteq R, R = \text{IDEAL}(q, \mathcal{D})], \quad (3.2)$$

where $\text{IDEAL}(\cdot)$ denotes the full pipeline described below.

3.2 Dual-Source Standpoint Generation

We propose that **forcing the model to express independent viewpoints about the answer**—one grounded in internal memory, the other in retrieved text—creates an explicit framework on which reconciliation can later operate.

3.2.1 Answer-Seen Standpoints (Seed Construction)

We sample a small seed set $\mathcal{C}_{\text{seed}}$ and reveal the answer a so that the model can observe an “ideal” reasoning target.

1. Internal knowledge elicitation

Following related studies [53, 43], we provide a rigorous instruction template to extract internal knowledge(K_{int}) from a frozen model using structured prompts 3.1.

$$K_{\text{int}}^* = \mathcal{E}_{\text{int}}(q; \Theta_0). \quad (3.3)$$

2. Independent standpoint construction

To prevent premature fusion (which could cause either source to discard useful information), we ask the model to articulate separate chains of reasoning under the condition that the gold answer is already known. Concretely, using distinct prompts (Prompt 3.2, 3.3) we generate:

$$\mathcal{S}_{\text{int}}^* = \mathcal{G}(q, a, K_{\text{int}}^*; \Theta_0), \quad \mathcal{S}_{\text{ext}}^* = \mathcal{G}(q, a, \mathcal{D}; \Theta_0). \quad (3.4)$$

By insisting that each source argues from its own evidence while the answer is visible, we capture two fully-developed, non-interfering lines of reasoning that will later be reconciled.

3. Exemplar bank population

$$\mathcal{B}_{\text{int}} \leftarrow \mathcal{S}_{\text{int}}^*, \quad \mathcal{B}_{\text{ext}} \leftarrow \mathcal{S}_{\text{ext}}^* \quad (3.5)$$

These high-fidelity cases serve as few-shot demonstrations for the answer-unseen stage.

Input:

Generate a document that provides accurate and relevant background knowledge related to the given question. The document should be informative and structured as if it were an excerpt from a knowledge source, without explicitly answering the question. Avoid unnecessary commentary, explanations, or direct responses. If relevant information is unavailable, state 'I don't know' without adding further speculation or context.

Question: {question}

Document: { K_{int} }

Prompt 3.1: Prompts to extract internal knowledge from a frozen model

3.2.2 Answer-Unseen Standpoints (Few-Shot Inference)

For every remaining instance, the answer is hidden. Like the prior section 3.2.1, the model first extracts fresh parametric facts

$$K_{\text{int}} = \mathcal{E}_{\text{int}}(q; \Theta_0), \quad (3.6)$$

Input:

Read the following documents relevant to the given question: $\{question\}$
 $\{retrieved\ documents\}$

Please answer the following question using external documents only, without relying on internal or prior knowledge: $\{question\}$ and explain how the proposed answer(s): $\{answers\}$ can be supported.

You are provided with a set of external documents. Base your explanation on the information found in these documents. If the documents do not reasonably support the proposed answer(s), you may instead present a more plausible answer that is supported by the documents, and explain why it fits better.

Do not refer to internal knowledge or prior facts beyond what the documents state or imply.

Your output should:

- Identify relevant factual claims from the external documents,
- Explain how those claims lead to the proposed answer, or support a better one, in a logically sound and document-grounded way.

Note that the question may be compositional and require intermediate analysis to deduce the final answer. Make sure your response is grounded and provides clear reasoning details followed by a concise conclusion.

Output: $\{S_{ext}^*\}$

Prompt 3.2: Prompts to generate external standpoint in answer-seen scenario.

Input:

Read the following documents relevant to the given question: $\{question\}$

$\{extracted\ internal\ knowledge\}$

Please answer the following question using internal knowledge only, without referring to any external documents: $\{question\}$ and explain how the proposed answer(s): $\{answers\}$ can be supported.

You are provided with a set of internal knowledge statements. Base your explanation primarily on these statements. If the internal knowledge does not reasonably support the proposed answer(s), you may instead present a more plausible answer that is supported by the internal knowledge, and explain why it fits better.

Your output should:

- Identify the factual claims from internal knowledge (provided or known),
- Explain how those claims lead to the proposed answer, or support a better one, in a logically sound and verifiable way.

Note that the question may be compositional and require intermediate analysis to deduce the final answer. Make sure your response is grounded and provides clear reasoning details, followed by a concise conclusion.

Output: $\{\mathcal{S}_{int}^*\}$

Prompt 3.3: Prompts to generate internal standpoint in answer-seen scenario.

then, conditioned on the exemplar banks, the model generates $\hat{\mathcal{S}}_{int}, \hat{\mathcal{S}}_{ext}$ using ICL Prompts 3.4 and 3.5, respectively.

$$\hat{\mathcal{S}}_{int} = \mathcal{G}(q, K_{int}; \Theta_0, ICE = \mathcal{B}_{int}), \quad \hat{\mathcal{S}}_{ext} = \mathcal{G}(q, \mathcal{D}; \Theta_0, ICE = \mathcal{B}_{ext}). \quad (3.7)$$

Each standpoint contains core evidence, a concise reasoning chain, and uncertainty notes, laying the groundwork for later fusion.

3.3 Linked Rationale Generation

Given the pair $(\hat{\mathcal{S}}_{int}, \hat{\mathcal{S}}_{ext})$, the task becomes: produce a coherent, conflict-aware rationale that justifies the answer. We achieve this with a Linked Rationale Generator $\mathcal{L}$, again operating in a two-step “answer-seen $\rightarrow$ answer-unseen” paradigm, plus an optional fine-tuning stage.

3.3.1 Answer-Seen Linked Exemplars

Input:

Your primary task is to answer the given question by analyzing the provided external documents. You must evaluate their relevance, accuracy, and completeness in relation to the question. If the documents clearly support a specific answer, explain how they lead you to that answer. If the documents are incomplete, ambiguous, or conflicting, make your best judgment based only on what is found in the documents. Do not use internal or prior knowledge.

Below are some examples of how to give the rationale:

$\{\mathcal{B}_{\text{ext}}\}$

Now it is your turn to analyze the following documents and answer the given question.

$\{\text{retrieved documents}\}$

Based on the provided information, answer the question: $\{\text{question}\}$

Output: $\{\hat{\mathcal{S}}_{\text{ext}}\}$

Prompt 3.4: Prompts to generate external standpoint in answer-unseen ICL task.

Input:

Your primary task is to answer the given question by analyzing the internal knowledge provided. You must examine this information to determine whether it contains enough evidence to support a clear answer. If it does, explain how the internal knowledge leads to your answer. If it does not, use your broader internal knowledge to offer the most plausible answer you can, but do not use any external sources or documents.

Below are some examples of how to give the rationale:

$\{\mathcal{B}_{\text{int}}\}$

Now it is your turn to analyze the following internal knowledge and answer the given question.

$\{\text{extracted internal knowledge}\}$

Based on your internal knowledge and the provided information, answer the question:

$\{\text{question}\}$

Output: $\{\hat{\mathcal{S}}_{\text{int}}\}$

Prompt 3.5: Prompts to generate internal standpoint in answer-unseen ICL task.

For each seed example, the model is instructed to: “Cross-examine the internal and external standpoints, highlight agreements or conflicts, supplement missing facts, and deliver a final conclusion.” The detailed prompt follows Prompt 3.6

The model outputs

$$\mathcal{R}_{\text{link}}^* = \mathcal{L}(q, a, \mathcal{D}, K_{\text{int}}^*, \mathcal{B}_{\text{int}}, \mathcal{B}_{\text{ext}}; \Theta_0), \quad (3.8)$$

which is stored in a third bank $\mathcal{B}_{\text{link}}$.

Input:

Read the following documents relevant to the given question: $\{question\}$

$\{retrieved\ documents\}$

$\{extracted\ internal\ knowledge\}$

You are given the following:

Question: $\{question\}$ Correct answer(s): $\{answers\}$

Two independent arguments attempt to justify what they believe to be the correct answer.

External Standpoint: $\{\mathcal{B}_{\text{ext}}\}$

Internal Standpoint: $\{\mathcal{B}_{\text{int}}\}$

Your task is to analyze both arguments and determine how internal and external information can contribute to reasoning toward the correct answer. Rather than choosing a side, your goal is to organize the relevant reasoning from both sources, identify which parts align with the correct answer, and explain how the conclusion can be supported.

In your explanation:

- Identify the key claims made in each argument.
- Compare them against the correct answer.
- Accept claims that logically support the correct answer.
- Reject claims that are inconsistent, unsupported, or contradict the correct answer and explain why.
- Integrate useful information from both sides to construct a coherent, step-by-step explanation that leads to the correct answer.

Base your explanation only on the arguments and the known correct answer.

Note that the question may be compositional and require intermediate analysis to deduce the final answer. Make sure your response is grounded and provides clear reasoning details followed by a concise conclusion.

Output: $\{\mathcal{R}_{\text{link}}^*\}$

Prompt 3.6: Prompts to generate linked rationale in answer-seen scenario.

3.3.2 Few-Shot Linking at Test Time

Even without additional training, Linked Generator $\mathcal{L}$ is capable of performing linked reasoning on unseen test instances $(q, \mathcal{D}) \in \mathcal{C}_{\text{test}}$ by leveraging the linking behavior $\mathcal{B}_{\text{link}}$, as illustrated in Prompt 3.7:

$$\hat{\mathcal{R}}_{\text{link}} = \mathcal{L}\left((q, \mathcal{D}, \hat{\mathcal{S}}_{\text{int}}, \hat{\mathcal{S}}_{\text{ext}}) \in \mathcal{C}_{\text{test}}; \Theta_0, \text{ICE} = \mathcal{B}_{\text{link}}\right). \quad (3.9)$$

Input:

Your primary task is to answer the given question by analyzing two competing arguments, each supported by a different type of information: one by external documents, the other by internal knowledge. Each argument includes its own reasoning and its supporting source content.

You must carefully evaluate how well each argument uses its respective source to justify its answer. If one side clearly provides stronger evidence and more valid reasoning, explain why you find it more convincing. If both arguments are incomplete, ambiguous, or equally strong, make your best judgment based only on the information provided. Do not rely on outside or prior knowledge.

Below are some examples of how to give the rationale:

$\{B_{\text{link}}^*\}$

Now it is your turn to analyze the following materials and answer the given question.

$\{\text{retrieved documents}\}$

$\{\text{extracted internal knowledge}\}$

Two independent arguments attempt to justify what they believe to be the correct answer.

External Standpoint: $\{\hat{\mathcal{S}}_{\text{ext}}\}$

Internal Standpoint: $\{\hat{\mathcal{S}}_{\text{int}}\}$

Based on the provided arguments and their supporting information, answer the question:
 $\{\text{question}\}$

Output: $\{\hat{\mathcal{R}}_{\text{link}}\}$

Prompt 3.7: Prompts to generate linked rationale in answer-unseen ICL task.

3.3.3 Instruction Tuning with Linked Rationales

In the SFT variant, the pipeline is repurposed to produce training pairs $(q, \mathcal{D}) \in \mathcal{C}_{\text{train}}$, which are then processed via Section 3.3.1 to obtain $\mathcal{R}_{\text{train-link}}^*$. These linked rationales are subse-

quently used to fine-tune the backbone model.

$$\Theta_{\text{link}} = \arg \min_{\Theta} \sum_{\mathcal{R}^* \in \mathcal{B}_{\text{train-link}}^*} -\log p_{\Theta}(\mathcal{R}^* \mid (q, \mathcal{D}, \hat{\mathcal{S}}_{\text{int}}, \hat{\mathcal{S}}_{\text{ext}}) \in \mathcal{C}_{\text{train}}). \quad (3.10)$$

After replacing the original weights with Θ_{link} , we generate $\hat{\mathcal{R}}_{\text{link}}$ under the ICE-free setting, following the conclusions drawn in [45].

$$\hat{\mathcal{R}}_{\text{link}} = \mathcal{L}\left((q, \mathcal{D}, \hat{\mathcal{S}}_{\text{int}}, \hat{\mathcal{S}}_{\text{ext}}) \in \mathcal{C}_{\text{test}}; \Theta_{\text{link}}\right). \quad (3.11)$$

Empirically, supervised fine-tuning (SFT) offers additional improvements on large-scale corpora, whereas in-context learning (ICL) alone already achieves strong performance.



Chapter 4

Experiments

This section details the data sets, evaluation metrics, baselines, training regime, and implementation settings used to assess IDEAL-RAG.

Section 4.2, 4.3 and 4.4 reports the main experimental result, while Section 5 digs deeper into error patterns and ablation studies.

4.1 Experimental Setup

4.1.1 Open-Domain QA Benchmarks

We measure open-domain QA performance on four widely-used collections that differ in question style and reasoning depth: PopQA[29] (single-fact name recall), Natural Questions[24] (web and Wikipedia questions), TriviaQA[21] (factoid trivia) and 2WikiMultiHopQA[13] (two-hop Wikipedia chains). For every question we keep the top- k passages returned by a hybrid retriever that combines BM25[35], DPR[22] and Contriever[17] over the [22] Wikipedia snapshot. This setup follows the evaluation protocol used in recent studies [1, 44, 45] that enabling fair and consistent comparison. Table 4.1 reports Recall@ k , confirming that sizeable portions of the gold evidence are nevertheless missed—an authentic imperfect retrieval scenario.

4.1.2 Counter-All and Counter-Mix Test Sets

In real-world retrieval pipelines, the top- k documents that the retriever returns are almost never “clean.” Instead, they are the output of a latent noise operator $\mathcal{N}$ that injects three broad types of corruption:

- **Irrelevant hits:** passages that do not pertain to the question at all;

- **Partial matches:** passages that mention some—but not all—facets of the answer;
- **Adversarial edits:** passages whose wording is intentionally or unintentionally misleading.

Formally, for each gold document set $\mathcal{D}_i$ we observe a perturbed set

$$\tilde{\mathcal{D}}_i = \mathcal{N}(\mathcal{D}_i) = \{\text{irrelevant hits}\} \cup \{\text{partial matches}\} \cup \{\text{adversarial edits}\}. \quad (4.1)$$

While most public benchmarks contain few genuinely adversarial passages, real-world settings more frequently feature such noisy or misleading content. To better reflect these challenges, we construct two counterfactual test suites that stress-test QA systems under heavier corruption, following the methodology of [9].

Whenever a retrieved passage contains the exact gold answer a , we **replace that span by a semantically similar but wrong entity**. The labeling process is processed by GPT-4o[16]. (e.g. “Barack Obama” $\rightarrow$ “Michelle Obama”).

- **Counter-All** ($\tilde{\mathcal{D}}_{c_a}$): every answer-containing passage is overwritten.
- **Counter-Mix** ($\tilde{\mathcal{D}}_{c_m}$): for questions with at least two ground-truth answers present in the retrieved contexts, we overwrite 50% of those passages with counterfactual content while leaving the remainder unchanged.

Table 4.1 reports, for every dataset, the number of queries that satisfy the substitution criteria and thus appear in $\tilde{\mathcal{D}}_{c_a}$ and $\tilde{\mathcal{D}}_{c_m}$. All downstream accuracy analyses are restricted to this subset to ensure comparability.

Dataset	Train	Test	Retriever	Top-K	R@K	$\tilde{\mathcal{D}}_{c_m}$	$\tilde{\mathcal{D}}_{c_a}$
PopQA	12,868	1,399	Contriever	5	68.7	578	961
Natural Questions	79,168	3,610	DPR	5	68.8	1,634	2,482
TriviaQA	78,785	11,313	Contriever	5	73.5	6,548	8,313
2WikiMultiHopQA	167,454	12,576	BM25	10	40.7	3,645	5,122

Table 4.1: Dataset statistics and retrieval setting (with counter values).

4.1.3 Evaluation Metrics

To make our results directly comparable with prior work [45, 1], we adopt the same exact-match accuracy (EM) metric as defined in Equation 3.2. A prediction is counted as correct when any string in the reference answer set is included in the model’s final answer span.

Accuracy Degradation Ratio (ADR). Robustness in heavily corrupted retrieval regimes is not captured by EM alone. We therefore introduce the Accuracy Degradation Ratio:

$$\text{ADR}_x = \frac{\text{EM}_{\text{clean}} - \text{EM}_{cf}}{\text{EM}_{\text{clean}}} \times 100\% \downarrow. \quad (4.2)$$

reports the percentage of accuracy lost when passages are replaced by the counterfactual variant $cf \in \{\tilde{\mathcal{D}}_{c_a}, \tilde{\mathcal{D}}_{c_m}\}$; lower is better.

Counterfactual Sensitivity Score (CSS). Beyond surface-level accuracy, we ask whether the model’s internal decision process is destabilised by adversarial evidence. We measure how strongly answer-token log-probabilities react to counterfactual edits:

$$\Delta\text{CSS} = \sum_{t \in \text{Ans}} \left| \log p_{\text{clean}}(t) - \log p_{cf}(t) \right| \downarrow. \quad (4.3)$$

The counterfactual sensitivity score sums the change in answer-token log-probabilities between the clean and edited contexts. A large value means the model’s belief swings sharply as soon as the passages are manipulated.

Parametric Knowledge Score (PKS). Following the analysis of [38], each transformer block integrates two distinct information flows: (i) a residual stream, which encapsulates the contextual information accumulated from the prompt (including any retrieved passages), and (ii) a feed-forward network (FFN), which introduces new evidence stored in the model’s parameters. PKS quantifies the extent to which the FFN reshapes the token-level logit distribution initially generated by the residual stream. For layer ℓ and answer token t :

$$\text{PKS}_{\ell,t} = \text{JSD}(\text{softmax}(W_U \text{LN}(h_{\ell,t}^{\text{mid}})), \text{softmax}(W_U \text{LN}(h_{\ell,t}^{\text{out}}))), \quad (4.4)$$

where h^{mid} is the hidden state *before* the FFN and h^{out} is *after* adding the FFN output back to the residual path.

- A **small PKS** means the FFN leaves the logits almost unchanged, so the block is essentially passing through the context it has already collected from the prompt.
- A **large PKS** means the FFN substantially *overwrites* those logits, signalling that the model is drawing on its *parametric memory* (facts encoded in the weights) rather than on the retrieved text.

For robustness analysis, we then report the per-layer change between clean and noisy contexts.

$$\Delta\text{PKS}_\ell = \frac{1}{|\text{Ans}|} \sum_{t \in \text{Ans}} (\text{PKS}_{\ell,t}^{\text{cf}} - \text{PKS}_{\ell,t}^{\text{clean}}), \quad (4.5)$$

A **positive** ΔPKS_ℓ therefore indicates that, under retrieval corruption, layer ℓ must compensate by injecting more stored knowledge, whereas a value close to zero signals that the model’s reliance pattern remains steady. Plotting the curve $\{\Delta\text{PKS}_\ell\}_{\ell=1}^L$ thus offers a fine-grained, layer-by-layer view of how noise shifts the balance between external evidence and internal memory.

All significance tests are conducted using a paired two-tailed t-test with a significance threshold of $p < 0.05$.

4.1.4 Baselines

Because *state-of-the-art* status in retrieval-augmented generation (RAG) depends heavily on domain, dataset size, and compute budget, it is difficult to nominate a single uncontested champion. We therefore evaluate IDEAL-RAG against a set of representative and well-performing systems that operate under experimental conditions close to ours and that span both **training-free** and **trainable paradigms**.

- **In-Context Retrieval-Augmented Language Modeling (RALM)[33]**. A training-free baseline that simply concatenates the top-k retrieved passages with the query inside the prompt. All reasoning is delegated to the frozen language model.
- **Vanilla Supervised Fine-Tuning (SFT)**. The most basic trainable baseline: given the retrieved passages, the model is fine-tuned to maximise the answer log-likelihood,

without any specialised prompting or auxiliary reasoning objectives.

- **InstructRAG[45]**. A trainable system that leverages few-shot exemplars to teach the model how to weave retrieved evidence into an explicit rationale. Among publicly reported RAG methods, it consistently achieves **strong denoising performance across open-domain QA benchmarks**.

For the sake of fairness, performance numbers marked with $\star$ are the best results we observe either in the authors’ original release (for InstructRAG) or in our own faithful re-implementation, whichever is higher. Note, however, that the original InstructRAG models were trained with **full-parameter fine-tuning**, whereas all of our experiments—including the IDEAL-RAG variants—employ **parameter-efficient fine-tuning (PEFT)**[12].

4.1.5 Implementation Details

All models are built on LLAMA-3-8B-Instruct. Fine-tuning uses LoRA[15] (rank 8, α 16, dropout 0.05) with two epochs, cosine-decayed AdamW[28] at 2.5×10^{-5} , warm-up 3 %, and a global batch of one million tokens accumulated on two A100-80 GB GPUs (DeepSpeed ZeRO-2[[32]], bf16). We retain just ten thousand randomly chosen training questions per set because the loss plateaus quickly. Inference is done with vLLM[25] in greedy mode; following the InstructRAG recipe, we include two exemplars per prompt when ICL is required.

4.2 Main Results

Table 4.2 reports exact-match (EM) accuracy on the four QA benchmarks under three retrieval settings: the original clean corpus and the two counterfactual test suites introduced in Section 4.1.2— $\tilde{\mathcal{D}}_{c_m}$ and $\tilde{\mathcal{D}}_{c_a}$. We summarise the key findings below.

4.2.1 Performance on Clean Data

We begin by evaluating Exact Match (EM) on the original, unperturbed corpus (Table 4.2). Across both training-free (prompt-only) and trainable (PEFT) configurations, IDEAL-RAG trails InstructRAG by a small margin. The largest trained-setting gap appears on 2Wiki-MultiHopQA (−7.2%), while in the training-free setting the biggest shortfall is on Natural

Method	PopQA			NaturalQuestions			TriviaQA			MultiHopQA		
	Original	$\tilde{\mathcal{D}}_{c_m}$	$\tilde{\mathcal{D}}_{c_a}$	Original	$\tilde{\mathcal{D}}_{c_m}$	$\tilde{\mathcal{D}}_{c_a}$	Original	$\tilde{\mathcal{D}}_{c_m}$	$\tilde{\mathcal{D}}_{c_a}$	Original	$\tilde{\mathcal{D}}_{c_m}$	$\tilde{\mathcal{D}}_{c_a}$
<i>w/o Training</i>												
RALM	61.97	72.66	32.36	56.37	69.34	25.38	71.47	82.48	42.80	43.37	60.81	54.98
InstructRAG	63.97	78.55	40.69	62.52	77.36	25.10	76.95	89.80	53.69	49.27	79.75	59.59
IDEAL-RAG	62.76	81.14	46.51	60.83	77.97	51.97	76.82	91.95	78.73	47.60	76.76	60.82
<i>w/ Training</i>												
vanilla*	61.00	—	—	56.60	—	—	73.90	—	—	43.8	—	—
InstructRAG*	65.90	89.45	49.32	65.68	80.29	32.67	78.70	90.79	57.80	57.19	87.02	66.26
IDEAL-RAG	64.05	84.08	47.76	63.71	80.97	55.48	77.19	92.21	78.58	50.01	80.85	64.31

Table 4.2: Exact-match accuracy (%) on four QA benchmarks. Columns report clean retrieval results (Original) as well as performance under two counterfactual corruption settings—50% passage edits ($\tilde{\mathcal{D}}_{c_m}$) and full-passage edits ($\tilde{\mathcal{D}}_{c_a}$). The upper block shows prompt-only models; the lower block includes models further fine-tuned with LoRA. *vanilla** scores are taken from [45], while InstructRAG* refers to the model checkpoint released by the same work, which yields stronger performance. Best numbers per column are **bold**.

Questions (−1.69%). As discussed in Section 5.1.2, IDEAL-RAG’s clean-set performance correlates with the availability of relevant parametric knowledge in the base model; consequently, benchmarks whose answers less frequently reside in the model’s memory—such as PopQA and 2WikiMultiHopQA—show the largest deficits relative to InstructRAG.

Importantly, these differences coexist with overall strong performance. When viewed against the retrieval upper bound, i.e., $R@k$ of answer-containing passages (reported in Table 4.1), IDEAL-RAG consistently operates near the empirical ceiling imposed by imperfect retrieval. In settings where $R@k$ is high, the gap to InstructRAG narrows; where $R@k$ is modest, both systems are similarly constrained by missing evidence. Notably, because IDEAL-RAG can fall back on parametric memory, its EM can occasionally meet or even exceed $R@k$ —a regime unattainable for methods that rely solely on retrieved passages. This perspective underscores that, even on clean data, IDEAL-RAG remains competitive while adhering to the practical limits of real-world retrieval.

4.2.2 Robustness under Counterfactual Noise

Under the counter-mix setting ($\tilde{\mathcal{D}}_{c_m}$), IDEAL-RAG attains parity with—or marginally surpasses—the strongest baselines, indicating that the introduction of even partial adversarial content is sufficient to expose the latter’s excessive dependence on retrieved passages. When

all answer-containing passages are counterfactually corrupted ($\tilde{\mathcal{D}}_{c_a}$), IDEAL-RAG demonstrates the greatest robustness, achieving a maximum improvement of +22.8% on Natural Questions; in the training-free configuration, this advantage widens to +26.9%. Comparable, although smaller, gains are observed on PopQA, TriviaQA, and 2WikiMultiHopQA.

A significant qualification emerges on benchmarks whose gold answers rarely reside in the model’s parametric memory—most notably PopQA and 2WikiMultiHopQA. In these cases, the *trained* InstructRAG baseline occasionally exceeds IDEAL-RAG under full corruption. We ascribe this to IDEAL-RAG’s deliberate emphasis on internally extracted evidence: when the parametric store offers little or no relevant information, that prior may become a liability. Section 5.1.2 provides an answer-containment analysis confirming this pattern: IDEAL-RAG dominates when the answer is available internally or jointly, yet trails the strongest baseline by a small margin when the answer resides exclusively in the retrieval.

Taken together, the findings substantiate our central claim: explicitly surfacing and aligning internal knowledge confers substantial resilience to retrieval noise, while highlighting the practical necessity of broad parametric coverage for domains where external evidence is the sole reliable source.

4.3 Accuracy Degradation Ratio (ADR)

Table 4.3 reports the Answer Degradation Rate (ADR)—the fraction of originally correct answers (on the clean corpus) that the model fails to retain once the retrieval context is corrupted. A **lower ADR** indicates **greater robustness**.

Empirical results across four QA benchmarks reveal that IDEAL-RAG achieves the lowest ADR under both $\tilde{\mathcal{D}}_{c_m}$ and $\tilde{\mathcal{D}}_{c_a}$ settings, indicating strong resistance to retrieval corruption, with wide margins in most settings. The contrast is most pronounced on the Natural Questions benchmark: in the rigorous $\tilde{\mathcal{D}}_{c_a}$ adversarial setting, InstructRAG suffers a 70.61% decline, while IDEAL-RAG registers a more modest 35.22 %—a near two-fold reduction in error amplification.

Similar—though less dramatic—improvements are observed on PopQA, TriviaQA, and 2WikiMultiHopQA, confirming that our explicit alignment of internal and external knowledge sub-

Method	PopQA		NaturalQuestions		TriviaQA		MultiHopQA	
	$\text{ADR}(\tilde{\mathcal{D}}_{c_m})$	$\text{ADR}(\tilde{\mathcal{D}}_{c_a})$	$\text{ADR}(\tilde{\mathcal{D}}_{c_m})$	$\text{ADR}(\tilde{\mathcal{D}}_{c_a})$	$\text{ADR}(\tilde{\mathcal{D}}_{c_m})$	$\text{ADR}(\tilde{\mathcal{D}}_{c_a})$	$\text{ADR}(\tilde{\mathcal{D}}_{c_m})$	$\text{ADR}(\tilde{\mathcal{D}}_{c_a})$
<i>w/o Training</i>								
RALM	20.46	63.33	18.83	67.83	13.11	53.16	4.49	21.47
InstructRAG	16.84	54.95	15.05	70.61	8.26	43.6	6.26	22.71
IDEAL-RAG	13.95	47.35	10.60	35.22	4.49	15.74	4.44	15.15
<i>w/ Training</i>								
InstructRAG*	11.53	50.68	14.64	63.15	7.34	39.77	4.03	21.84
IDEAL-RAG	11.64	46.76	9.56	33.64	4.64	16.27	4.50	15.67

Table 4.3: ADR ($\downarrow$) measures accuracy drop from clean to corrupted retrieval. We compare methods on four QA benchmarks under partial ($\tilde{\mathcal{D}}_{c_m}$) and full ($\tilde{\mathcal{D}}_{c_a}$) passage corruption. IDEAL-RAG consistently shows the lowest degradation, both with and without fine-tuning.

stantially improves model robustness, even when all retrieved passages are misleading.

Overall, IDEAL-RAG not only narrows the accuracy gap in noisy settings (Section 4.2) but also retains a far greater share of its clean-data competence, **underscoring its practical reliability in real-world retrieval scenarios.**

4.4 Internal-Mechanism Analysis

4.4.1 Counterfactual Sensitivity Score

For each question–answer pair we measure the drop, shown in Equation 4.3 in the answer logit after replacing the clean retrieval set with a corrupted one.

We plot the empirical distribution of ΔCSS as a violin plot; a “fatter” violin (larger inter-quartile width) indicates that retrieval noise exerts a stronger—and more variable—effect on the model’s confidence. The result is shown in Figure 4.1.

- IDEAL-RAG exhibits markedly narrower violins on Natural Questions and TriviaQA, signalling that its answer probabilities remain tightly concentrated even when every passage is adversarial.
- InstructRAG, by contrast, shows broad, heavy-tailed violins, reflecting its heavier reliance on the surface form of the retrieved context.

For example, the average ΔCSS on Natural Questions under the fully adversarial $\tilde{\mathcal{D}}_{c_a}$ set-

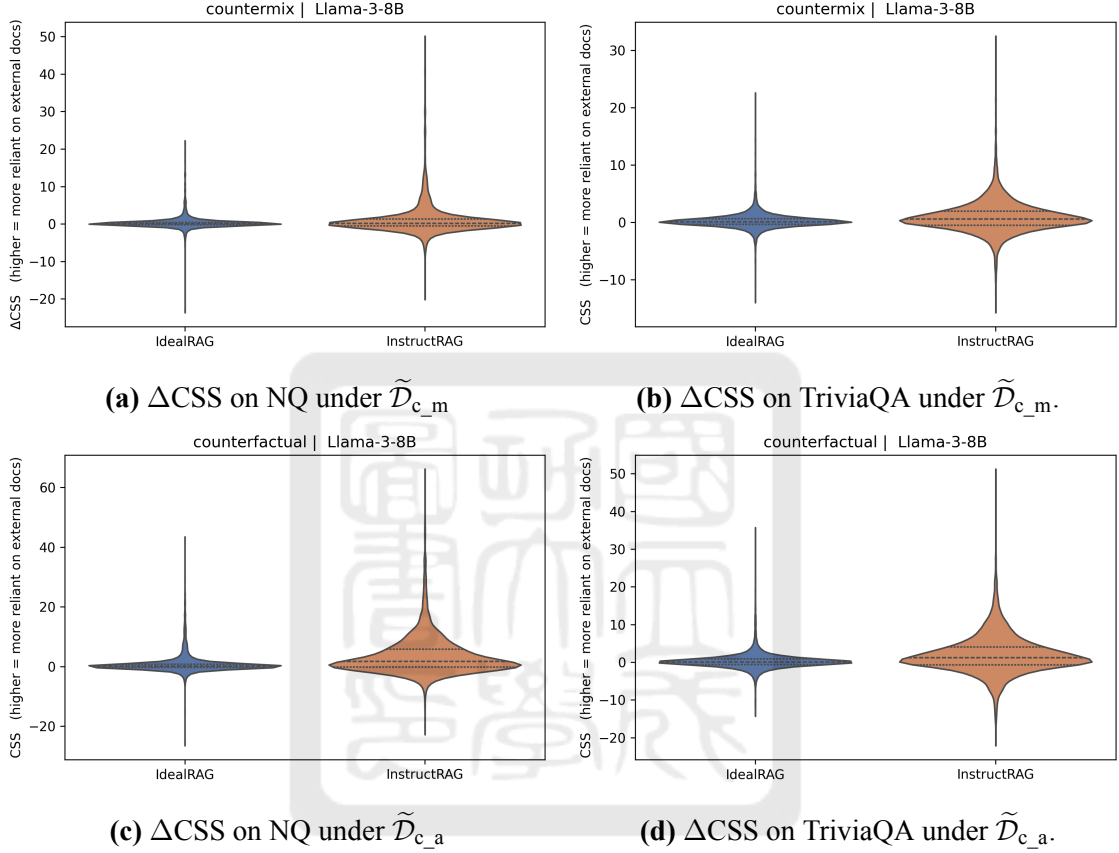


Figure 4.1: Violin plots compare the change in answer-token log-probabilities between the clean retrieval set and (a, b) the mixed-counterfactual set $\tilde{\mathcal{D}}_{c_m}$ or (c, d) the fully counterfactual set $\tilde{\mathcal{D}}_{c_a}$ on Natural Questions and TriviaQA. Narrower, lower-centred violins (blue, IDEAL-RAG) indicate that the model’s confidence is largely unchanged when passages are edited, whereas wider, higher-centred violins (orange, InstructRAG) reveal a stronger and more volatile dependence on the corrupted evidence.

ting is 0.830 for IDEAL-RAG versus 3.657 for InstructRAG. This substantial difference reinforces the qualitative trend highlighted by the ADR analysis (Section 4.3): IDEAL-RAG’s decision boundary is significantly more stable and less susceptible to misleading evidence.

4.4.2 Parametric Knowledge Score

We measure how much each transformer block must inject parametric knowledge once retrieval becomes unreliable. Following Equation 4.5, we compute the layer-wise change ΔPKS_ℓ between the clean context and its corrupted counterpart, as shown in Figure 4.2.

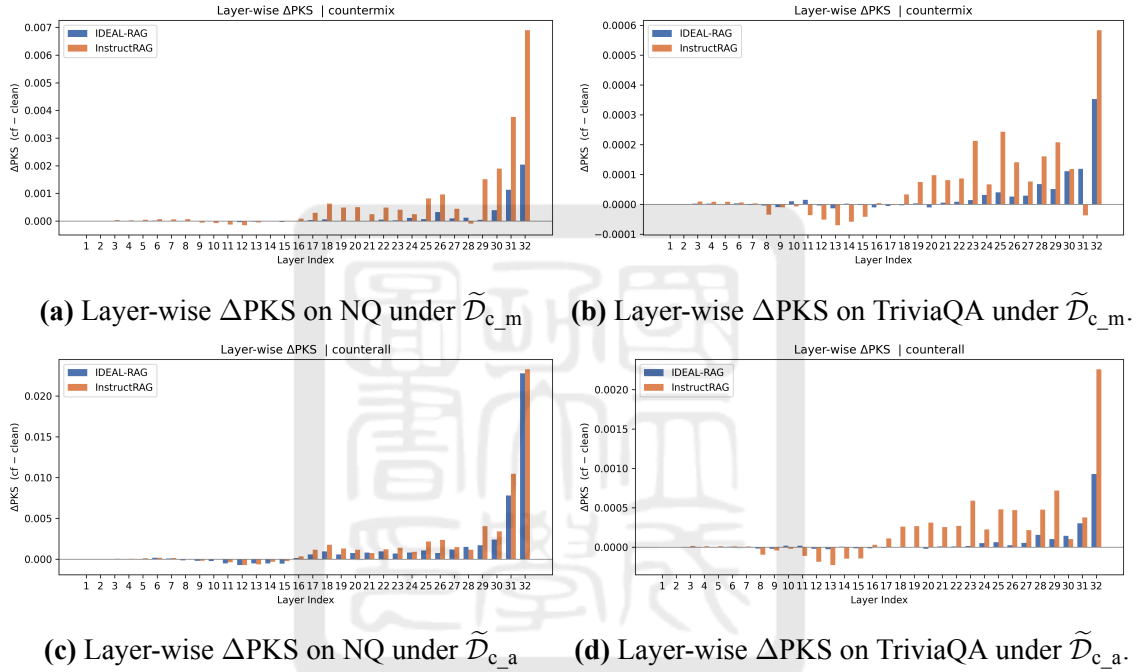


Figure 4.2: For each transformer block we plot the rise in PKS after replacing the clean passages with counterfactual ones. Orange bars are IDEAL-RAG, blue bars InstructRAG. (a, b) show the 50 % corruption mix ($\tilde{\mathcal{D}}_{c_m}$) on Natural Questions and TriviaQA; (c, d) show the full-corruption setting ($\tilde{\mathcal{D}}_{c_a}$). IDEAL-RAG requires only modest additional parametric input at deeper layers and maintains stable behavior, whereas InstructRAG exhibits larger spikes, indicating a late, reactive fallback to internal memory when retrieval is unreliable.

Aggregate comparison. On the two benchmarks we analyse—Natural Questions and TriviaQA—both IDEAL-RAG and InstructRAG display higher ΔPKS under noise than under clean retrieval, indicating that LLMs fall back on stored memory when external evidence degrades. Crucially, IDEAL-RAG exhibits consistently lower ΔPKS compared to InstructRAG. This is because IDEAL-RAG surfaces relevant internal knowledge early in the extraction stage, reducing the need for additional information injection through the knowledge-FFN

path. As a result, the residual stream remains more stable, mitigating the risk of hallucinations. This observation aligns with the findings from REDEEP[38], which associate excessive late-stage activation with increased generation instability.

Outcome-conditioned analysis. To move beyond aggregate correlations, we partition IDEAL-RAG’s test predictions into **Correct** and **Incorrect** subsets. The results in Figure 4.3 show that $\Delta\text{PKS}^{\text{wrong}}$ is substantially higher—often exceeding the value of $\Delta\text{PKS}^{\text{correct}}$. This indicates that IDEAL-RAG’s remaining errors tend to arise when the model abruptly amplifies its knowledge-FFN contribution to compensate for misleading retrieval. In contrast, predictions with small ΔPKS are consistently reliable, suggesting greater stability under corrupted input.

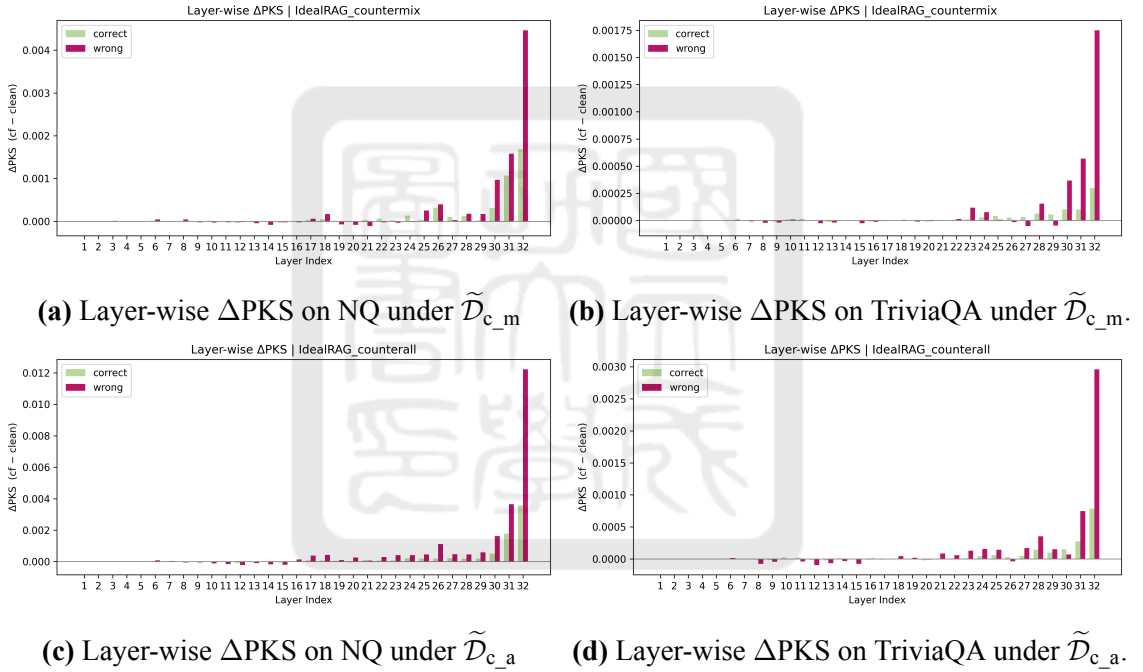
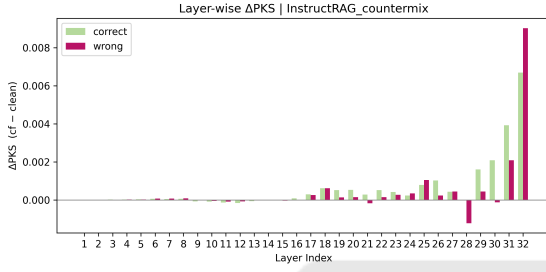


Figure 4.3: (green = answers correct, red = answers wrong) further shows that within IDEAL-RAG itself, questions it fails on are accompanied by a sharper late-layer ΔPKS surge—especially in the final two blocks—whereas successful cases keep the rise modest. Hence a sudden, large jump in parametric logits is a reliable warning signal for impending hallucination under both the 50 %-mix and full-corruption settings on Natural Questions and TriviaQA. (Layout and subplot lettering follow Fig 4.2 for visual consistency.)

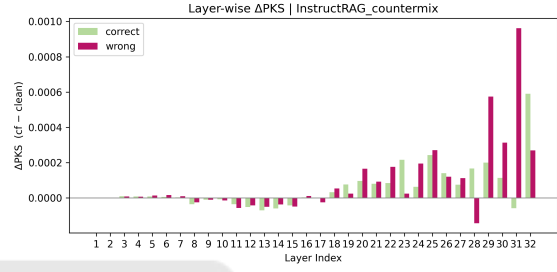
Outcome-conditioned ΔPKS for InstructRAG. The results are shown in Figure 4.4. Unlike IDEAL-RAG—where incorrect predictions reliably exhibit larger layer-wise ΔPKS than correct ones—InstructRAG shows little discriminative structure. On both Natural Questions and TriviaQA, the ΔPKS profiles of Correct vs. Incorrect cases largely overlap and fluctu-

ate across the deepest blocks, with sharp spikes appearing in both outcomes. This pattern is consistent with a reactive, uncalibrated fallback to parametric memory: because no internal-knowledge extraction or cross-source reconciliation is performed, the model intermittently amplifies the knowledge-FFN pathway when retrieval becomes unreliable. Whether this late amplification helps or hurts depends idiosyncratically on each instance (e.g., whether the memorized fact is accurate and salient), yielding near-random association between ΔPKS magnitude and accuracy. these observations indicate that ΔPKS functions as a reliable error signal only when the pipeline explicitly surfaces and structures parametric evidence—as in IDEAL-RAG. Under the one-shot instruction regime of InstructRAG, ΔPKS mainly reflects unstable residual competition rather than principled integration, which is consistent with REDEEP’ s observation that uncontrolled late knowledge-FFN dominance correlates with hallucinations. This mechanistic interpretation also accords with the aggregate accuracy gaps in our main results: under both the counter-mix and counter-all settings ($\tilde{\mathcal{D}}_{\text{c_m}}$ and $\tilde{\mathcal{D}}_{\text{c_a}}$), IDEAL-RAG surpasses InstructRAG (see Table 4.2), mirroring the more stable ΔPKS profile observed for our method.

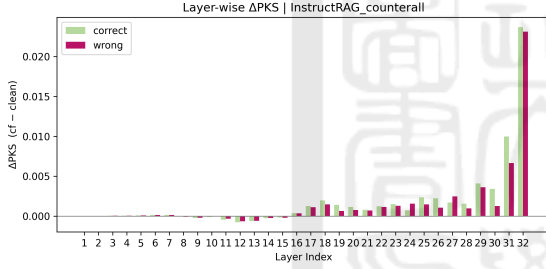
In summary, the outcome-conditioned analyses reveal that large, abrupt ΔPKS spikes are strong indicators of model error. By proactively extracting relevant internal knowledge, IDEAL-RAG maintains a balanced residual stream and prevents large ΔPKS spikes, thereby improving robustness to noisy retrieval. In contrast, InstructRAG reacts to corrupted inputs with abrupt increases in knowledge-FFN contribution, echoing REDEEP’ s finding that such reactive surges are strongly associated with hallucinations.



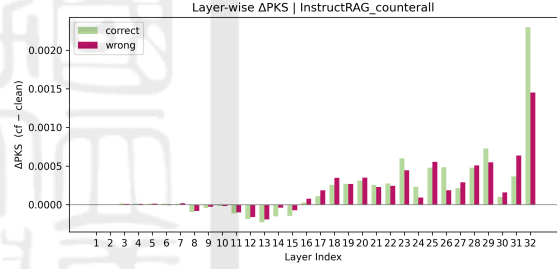
(a) Layer-wise ΔPKS on NQ under $\tilde{\mathcal{D}}_{c_m}$



(b) Layer-wise ΔPKS on TriviaQA under $\tilde{\mathcal{D}}_{c_m}$



(c) Layer-wise ΔPKS on NQ under $\tilde{\mathcal{D}}_{c_a}$



(d) Layer-wise ΔPKS on TriviaQA under $\tilde{\mathcal{D}}_{c_a}$

Figure 4.4: (green = answers correct, red = answers wrong) shows little separation between outcomes: both correct and wrong cases exhibit erratic, late-layer spikes with wide overlap and no stable trend, under both the 50%-mix and full-corruption settings on Natural Questions and TriviaQA. This pattern suggests ΔPKS is a weak diagnostic for this one-shot regime—reflecting reactive, unstable use of parametric memory rather than a structured fallback. (Layout and subplot lettering match Fig. 4.2.)

Chapter 5

Analysis

In this section we dissect IDEAL-RAG along three axes: (i) ablations that remove individual components (Section 5.1), (ii) training-data interventions that expose whether our robustness stems from architecture or data (Section 5.2), and (iii) an attention probe that visualises where the model looks when confronted with counterfactual evidence (Section 5.3).

5.1 Ablation Study

5.1.1 Framework Ablation

IDEAL-RAG proceeds in three stages—**Parametric Knowledge Extraction** ($\mathcal{E}_{\text{int}}$), **Dual-Source Standpoint Generation** ($\mathcal{G}_{\text{int}}, \mathcal{G}_{\text{ext}}$), and **Linked Rationale Generation** ($\mathcal{L}$).

We evaluate two reduced systems:

1. **w/o $\mathcal{E}_{\text{int}}$ and $\mathcal{G}$**

Here we replicate InstructRAG’s one-shot recipe: a single prompt (Prompt 5.1) asks the model to read the retrieved passages, reflect on any relevant background knowledge, and then justify its answer—yet it omits both the explicit parametric-knowledge extraction and the dual-standpoint stage. Apart from those two removals, the instruction style, exemplar count, decoding settings, and evidence-quality guidance are kept identical to IDEAL-RAG, so any performance gap can be attributed to the missing mechanisms rather than to prompt design.

2. **w/o $\mathcal{G}$**

We still extract parametric knowledge with $\mathcal{E}_{\text{int}}$ but skip the separate standpoint phase, feeding the question, passages, and extracted memory directly into the fusion prompt, as shown in Prompt 5.2.

Input:

Your primary task is to answer the given question by first reflecting on what internal knowledge you have that might be relevant, and then critically analyzing the provided documents. You must evaluate the relevance, accuracy, and sufficiency of both internal and external information in relation to the question.

Below are some examples of how to give the rationale:

$\{B_{\text{inst}}^*\}$

Now it is your turn to analyze the following documents and answer the given question.

{retrieved documents}

Based on both your internal knowledge and the provided information, answer the question:

Question: *{question}*

Output:

Prompt 5.1: Prompts that refine the InstructRAG setup to generate rationales in the answer-unseen ICL task. The exemplar bank B_{inst}^* contains rationales generated under the InstructRAG method in the answer-seen setting.

Input:

Your primary task is to answer the given question by analyzing both the provided external documents and internal knowledge. You must evaluate how well each source contributes to answering the question, and whether they support one or more of the proposed answer labels. Below are some examples of how to give the rationale:

$\{B_{\text{one-step}}^*\}$

Now it is your turn to analyze the following materials and answer the given question.

{retrieved documents}

Answer a given question using the information from both externally retrieved documents and your own memorized documents.

Question: *{question}*

Output:

Prompt 5.2: Prompts for generating linked rationales without explicit standpoints ($\mathcal{G}$) in the answer-unseen ICL setting.

Results on Natural Questions and TriviaQA (Table 5.1) reveal:

- Parametric extraction is decisive.** When we collapse IDEAL-RAG into a single one-shot prompt that neither extracts parametric knowledge nor separates standpoints, performance craters exactly under full counterfactual noise: on Natural Questions, EM drops from 51.97 (full model) to 30.74 (−21.23%), and on TriviaQA from 78.58 to 52.85 (−25.73%). By contrast, changes on the clean and counter-mix splits are small ($\pm 2\%$). This pattern isolates explicit parametric-knowledge extraction (and the accompanying dual-standpoint setup) as the core source of robustness: without it, the model is easily swayed by fully adversarial passages, even though surface accuracy on cleaner settings appears largely unchanged.
- Standpoint generation matter, but are complementary.** Removing the standpoint generation stage while keeping explicit parametric extraction yields a modest average EM drop of about 2–3% across clean and noisy splits. This confirms that cross-source reconciliation is beneficial, yet its contribution is smaller than that of the explicit parametric-knowledge step, whose removal causes far larger degradations. Case analyses further show that standpoint generation helps in edge scenarios—resolving residual conflicts, tempering spurious cues, and producing more consistent, interpretable justifications—but the primary driver of robustness remains the prior elicitation of internal knowledge. Accordingly, our deeper analysis emphasizes the explicit-extraction stage, viewing standpoint generation as a complementary stabilizer rather than the dominant factor.

Method	NaturalQuestions			TriviaQA		
	Original	$\tilde{\mathcal{D}}_{c_m}$	$\tilde{\mathcal{D}}_{c_a}$	Original	$\tilde{\mathcal{D}}_{c_m}$	$\tilde{\mathcal{D}}_{c_a}$
w/o $\mathcal{E}_{int}$ & $\mathcal{G}$	61.77 ($\uparrow 0.94\%$)	79.74 ($\uparrow 1.77\%$)	30.74 ($\downarrow 21.23\%$)	76.05 ($\downarrow 1.14\%$)	90.13 ($\downarrow 2.08\%$)	52.85 ($\downarrow 25.73\%$)
w/o $\mathcal{G}$	60.86 ($\uparrow 0.03\%$)	76.99 ($\downarrow 0.98\%$)	48.71 ($\downarrow 3.26\%$)	76.07 ($\downarrow 1.12\%$)	92.20 ($\downarrow 0.01\%$)	76.68 ($\downarrow 1.90\%$)
w/ all	<u>60.83</u>	<u>77.97</u>	<u>51.97</u>	<u>77.19</u>	<u>92.21</u>	<u>78.58</u>

Table 5.1: Columns report performance on the original retrieval context and on the two counterfactual test suites— $\tilde{\mathcal{D}}_{c_m}$ and $\tilde{\mathcal{D}}_{c_a}$. Rows progressively omit key IDEAL-RAG stages: (i) both parametric-knowledge extraction and standpoints, (ii) standpoints only, and (iii) the full model.

5.1.2 Answer-Containment Analysis

Mirroring the data construction in Section 4.1.2, we partition the clean test questions according to where the gold answer appears:

- **inter_only** ($\mathcal{D}_{\text{in}}$): the gold answer appears in K_{int} but not in the retrieved passages;
- **exter_only** ($\mathcal{D}_{\text{ex}}$): the opposite;
- **both_contained** ($\mathcal{D}_{\text{both}}$): both sources contain the answer.

The performance in Table 5.2 shows that IDEAL-RAG consistently outperforms InstructRAG when the model already knows the answer (inter-only) or when both internal and retrieved knowledge align. It yields a slight performance drop only when the answer resides exclusively in the retrieved passages. These results suggest that the explicit use of parametric memory provides the greatest benefit in precisely the scenarios for which it was designed.

Method	PopQA			NaturalQuestions			TriviaQA			MultiHopQA		
	$\mathcal{D}_{\text{in}}$	$\mathcal{D}_{\text{ex}}$	$\mathcal{D}_{\text{both}}$	$\mathcal{D}_{\text{in}}$	$\mathcal{D}_{\text{ex}}$	$\mathcal{D}_{\text{both}}$	$\mathcal{D}_{\text{in}}$	$\mathcal{D}_{\text{ex}}$	$\mathcal{D}_{\text{both}}$	$\mathcal{D}_{\text{in}}$	$\mathcal{D}_{\text{ex}}$	$\mathcal{D}_{\text{both}}$
<i>w/o Training</i>												
InstructRAG	36.36	86.76	94.54	35.42	76.25	91.80	59.28	89.27	96.69	78.44	53.57	90.21
IDEAL-RAG	54.55	82.34	95.45	71.88	70.07	93.03	81.01	78.66	97.21	86.11	36.88	91.06
<i>w/ Training</i>												
InstructRAG*	40.91	89.64	94.77	39.58	82.83	92.75	64.03	90.40	97.37	81.36	68.36	93.92
IDEAL-RAG	63.64	86.69	96.82	71.88	70.07	93.03	83.23	78.19	97.83	90.29	44.41	94.01

Table 5.2: For each datasets we report exact-match accuracy when the gold answer appears only in the model’s parametric memory ($\mathcal{D}_{\text{in}}$), only in the retrieved passages ($\mathcal{D}_{\text{ex}}$), or in both sources ($\mathcal{D}_{\text{both}}$).

5.2 Training Data Analysis

To verify that IDEAL-RAG’s improvements stem from its architecture rather than data alone, we fine-tune both IDEAL-RAG and InstructRAG on 5,000 counterfactual training instances from Natural Questions, constructed following Section 4.1.2. The results are presented in Table 5.3.

While the added noise enhances InstructRAG’s robustness, it **reduces** its clean-set accuracy. In contrast, IDEAL-RAG **improves on both clean and noisy sets simultaneously**, and

Method	Natural Questions					
	$Origin_{c_m}$	$\tilde{D}_{c_m}$	$ADR(\tilde{D}_{c_m})$	$Origin_{c_a}$	$\tilde{D}_{c_a}$	$ADR(\tilde{D}_{c_a})$
<i>Training w/ normal data</i>						
InstructRAG*	94.06	80.29	14.64	88.68	32.68	63.15
IDEAL-RAG	89.53	80.97	9.56	83.6	55.48	33.64
<i>Training w/ counter_mix data</i>						
InstructRAG	90.7	81.33	10.33	83.16	30.7	63.08
IDEAL-RAG	89.84	81.88	8.86	83.56	54.47	34.81
<i>Training w/ counter_all data</i>						
InstructRAG	90.7	83.11	8.37	83.48	37.27	55.35
IDEAL-RAG	90.02	82.86	7.95	83.96	55.56	33.83

Table 5.3: Natural-Questions results under three training regimes. Each block reports accuracy on the clean set (Original, which meets the counterfactual construction criteria) as well as on two counterfactual test sets ($\tilde{D}_{c_m}, \tilde{D}_{c_a}$). The corresponding Answer Degradation Rate (ADR) is also included. Instruct* is the released full-FT model; others use PEFT. IDEAL-RAG keeps comparable clean accuracy and yields much lower ADR than InstructRAG across all training data conditions.

still outperforms InstructRAG under noisy conditions—suggesting that the improvements are attributable to architectural advantages rather than data-specific effects.

5.3 Attention-Distribution Probe

To probe the model’s decision focus, we aggregate token-level self-attention across all decoder layers and heads, then sum the weights per retrieved passage p_j :

$$\text{AttnScore}(p_j) = \sum_{\ell, h} \sum_{t \in p_j} \text{softmax}(A_{\ell, h})_t, \quad (5.1)$$

where $A_{\ell, h}$ is the raw attention matrix at layer ℓ , head h .

Figure 5.1 plots these scores on a example from $\tilde{D}_{c_m}$ set:

InstructRAG concentrates almost exclusively on one or two passages, often ignoring the passage that actually contains the gold answer—a pattern that explains its high $\tilde{D}_{c_f}$ fragility. IDEAL-RAG distributes attention far more evenly and, crucially, assigns the largest weight to both the true-answer and the counterfactual passages—the two sources it must compare to

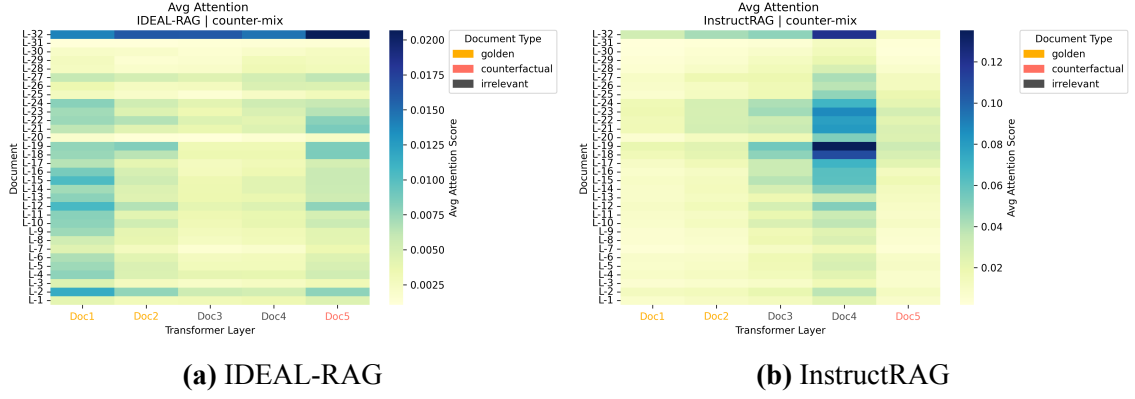


Figure 5.1: Passage-level attention heat-maps on a counter-mix example. IDEAL-RAG spreads attention across all passages and highlights both the gold-answer and counterfactual segments, whereas InstructRAG concentrates on a single irrelevant passage.

justify or refute conflicting claims.

This even-handed attention pattern aligns with the design intent of the Linked Rationale module and further corroborates IDEAL-RAG’ s resilience in noisy retrieval scenarios.

5.4 Summary of Analysis

The analyses confirm that each design choice—memory extraction, dual standpoints, and linking-based integration—contributes to the model’ s ability to resist misleading evidence, and that IDEAL-RAG’ s gains persist even when the training data distribution changes.

Chapter 6

Conclusion

This work revisits retrieval-augmented generation from the perspective that has so far received surprisingly little attention: **the language model’s own parametric memory**. We introduced **IDEAL-RAG**, a three-stage framework that (i) makes the model surface whatever it already “knows” about the query, (ii) lets internal and external sources each produce a **self-contained** line of reasoning, and (iii) then **mutually verifies** the two standpoints, combines overlapping clues, resolves mismatches when possible, and synthesises the answer that is most consistently supported.

Experiments across four open-domain QA benchmarks show that this division of labor preserves the strong performance of standard RAG systems under clean retrieval, while significantly mitigating accuracy degradation under partially or fully counterfactual passages. Mechanistic diagnostics support this: a lower CSS indicates answer probabilities are less perturbed by noisy passages, and layer-wise PKS analyses reveal steadier reliance on internal memory, with outcome-conditioned spikes serving as warning signals for errors. Ablations demonstrate that removing any of the three stages, especially the initial extraction of parametric knowledge, significantly contributes to this robustness.

In conclusion, IDEAL-RAG shows that explicitly modeling and aligning internal and external evidence provides an effective route to noise-resilient retrieval-augmented generation. **Our findings suggest that deliberate negotiation between what an LLM *knows* and what it *reads* offers a principled foundation for more dependable RAG systems.** Future work will extend this negotiation framework to longer contexts, adaptive retrieval, and multi-step reasoning tasks beyond question answering.

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